

Lecture 01

Introduction to Environmental Economics

Ivan Rudik

AEM 4510 / AEM 5510 / ECON 3865

Roadmap

- Course set up
- What is environmental economics?
- What are the goals for this class?
- Microeconomics recap

Course set up

Basics

Professor: Ivan Rudik

TAs: Chrisma D'Souza, Nathan Bol, Becca Ford

Office hours: see canvas

Readings: Markets and the Environment by Keohane and Olmstead, papers available through the course website or library

Prereqs: MATH 1110, and either AEM 2600 or ECON 3030

Grading

4 prelims (20% each):

- Feb 12, Mar 17, Apr 14, May 5
- Prelims will be a combination of multiple choice, short answer, graphical, and simple mathematical questions that will not require a calculator
- **SDS:** This class is in the Alternative Test Program, if approved for exam accommodations, you must request your exam accommodation(s) for this course and fill out an exam request form for each exam in this course via the SDS student portal by February 8th
- No makeup prelims: if you miss a prelim, it's weight is equally distributed to your other prelims

Grading

4 Problem Sets (5% each): mix of quantitative and conceptual problems to give you practice in learning key concepts (teams of 3)

- 1 before each prelim: Feb 10, Mar 12, Apr 9, Apr 30
- Solutions posted immediately after the deadline
- No late submissions

Content

Weeks 1-7ish: the economics of environmental regulation

- More theoretical, graphical
- Taxes, cap and trade, regulating monopolies, tariffing other countries, etc

Weeks 7-14ish: using markets (real and financial) to learn about the environment

- Some theory, also real world analyses and data
- valuing recreation, housing markets, bond markets, green instruments

In class

No recording or sharing of materials outside class: Each person in this class is expected to respect the principles of academic freedom for instructors and classmates and will maintain the privacy of the classroom environment, as outlined in Cornell's Code of Academic Integrity.

No use of electronics in class: the use of laptops, tablets, phones, or other electronics in class is prohibited. Notes must be taken on paper.

Why no electronics?

The reason for no electronics is pedagogical: **to improve learning in the course.**

Laptop or tablet use in class (even just for notes) is associated with decreased learning and lower test scores in a variety of settings

I post slides after we finish each section so you will not miss any content

Sana, F., Weston, T., & Cepeda, N. (2013). *Computers & Education*, 62, 24-31.; Mueller, P. & Oppenheimer, D. (2014). *Psychological Science*, 25(6), 1159-1168.; Carter, S. P., Greenberg, K., & Walker, M. S. (2017). *Economics of Education Review*, 56, 118-132.

Last words

This was just an overview, canvas has a lot more details about specific requirements

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Any questions about the course?

What is environmental economics?

What's enviro econ?

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Everything is scarce compared to wants and needs

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These ideas can be applied to the environment

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The value of mitigating pollution

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How agents will response to climate change policies

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Environmental economics helps us understand things like:

The value of mitigating pollution

How agents will response to climate change policies

Whether investment tax credits for wind power are cost-effective

Air pollution is bad



How do people respond to info?



ADVISORY: #ozone is expected to be Unhealthy for All today in and around the foothills area of @SequoiaKingsNPS (Ash Mountain entrance)



Ozone Air Quality Guide

Advisory Level	Health Concern
Unhealthy (86 ppb – 105 ppb)	The following groups should avoid prolonged outdoor exertion: • People with lung disease, such as asthma • Children and older adults • People who are active outdoors Everyone else should limit prolonged outdoor exertion.

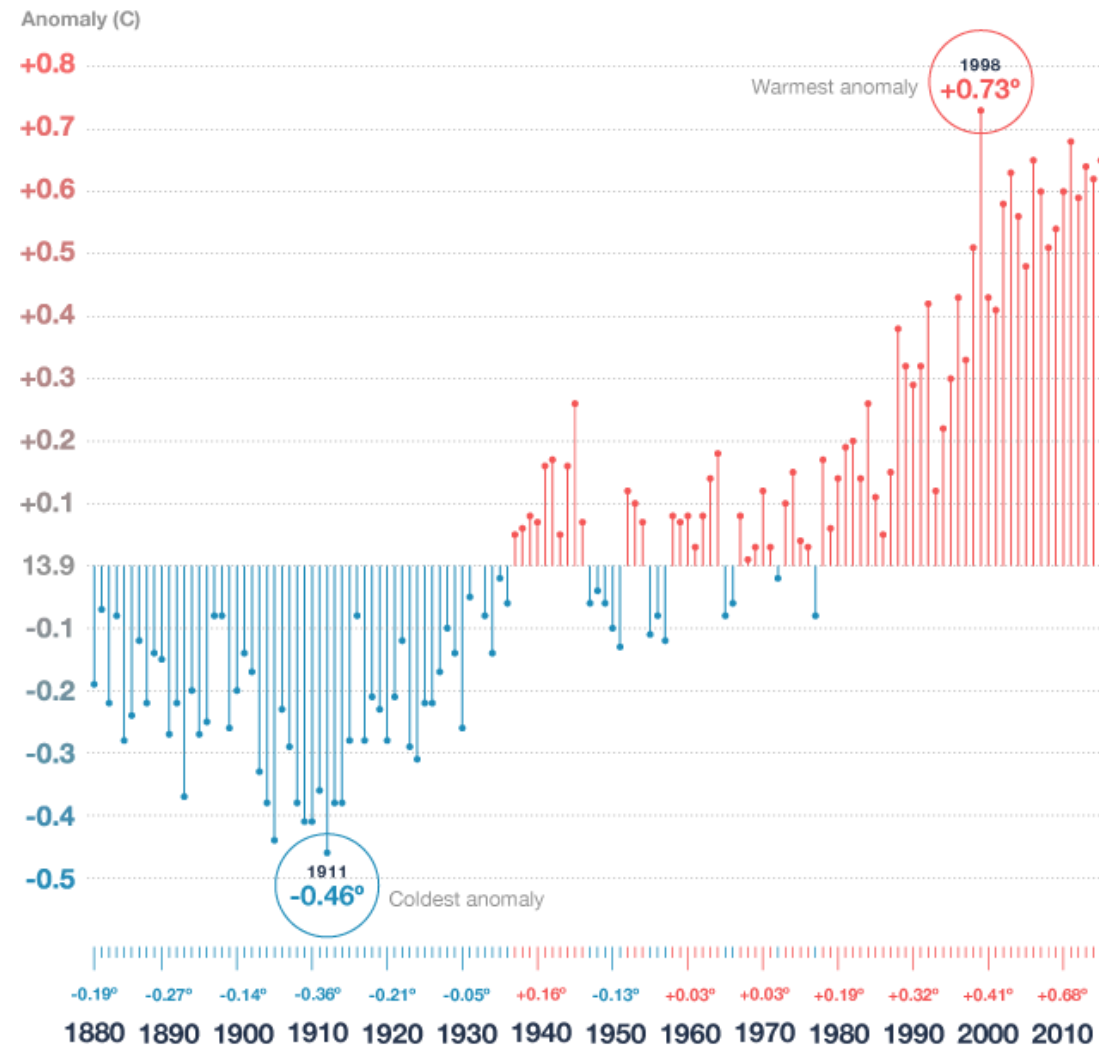
The ozone advisory level is based on an 8 hour average ozone concentration; ppb = parts per billion.

Places provide info to help people avoid air pollution

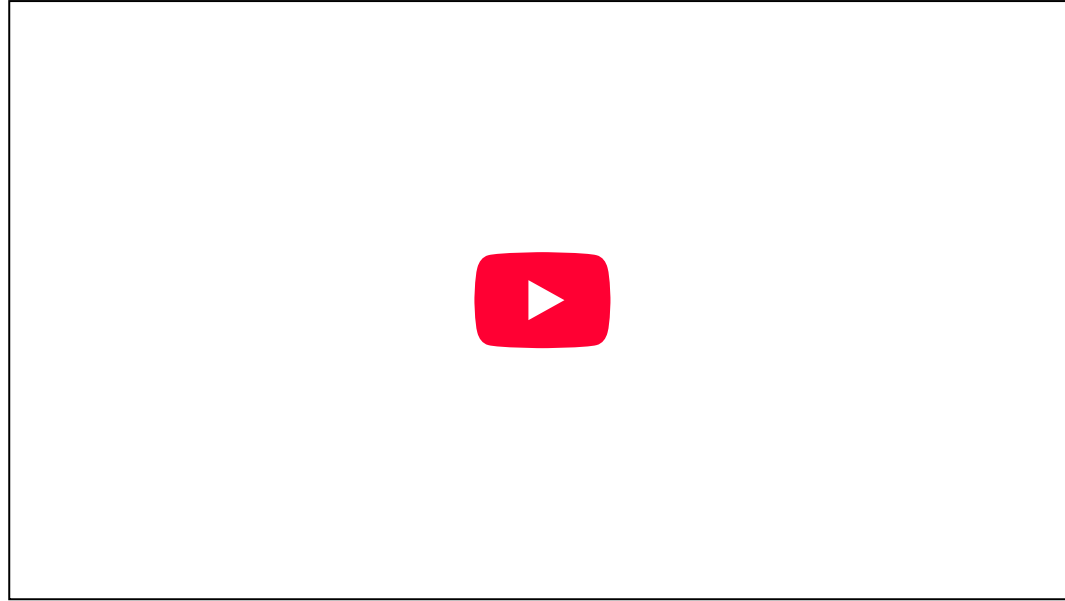
Does it work?

How well?

Climate change



Climate change



Climate change

Why do economists care about climate change?

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- Fishing

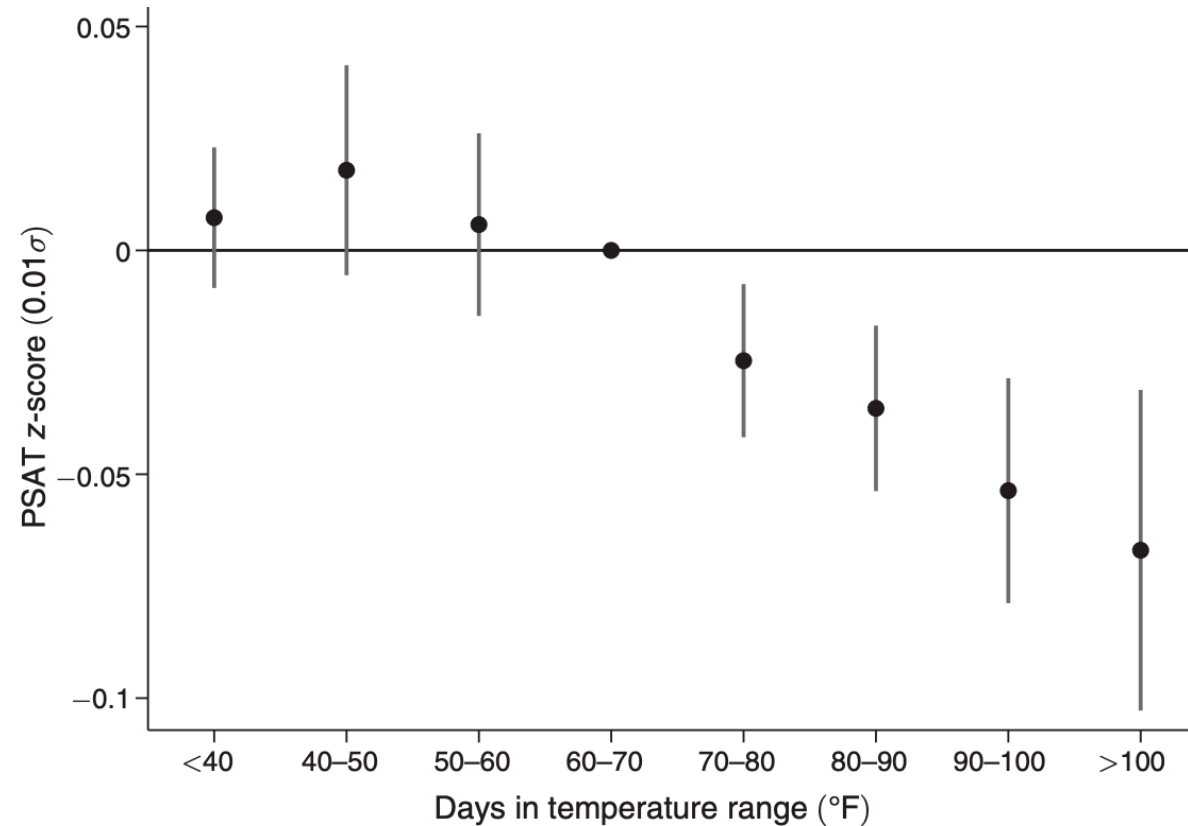
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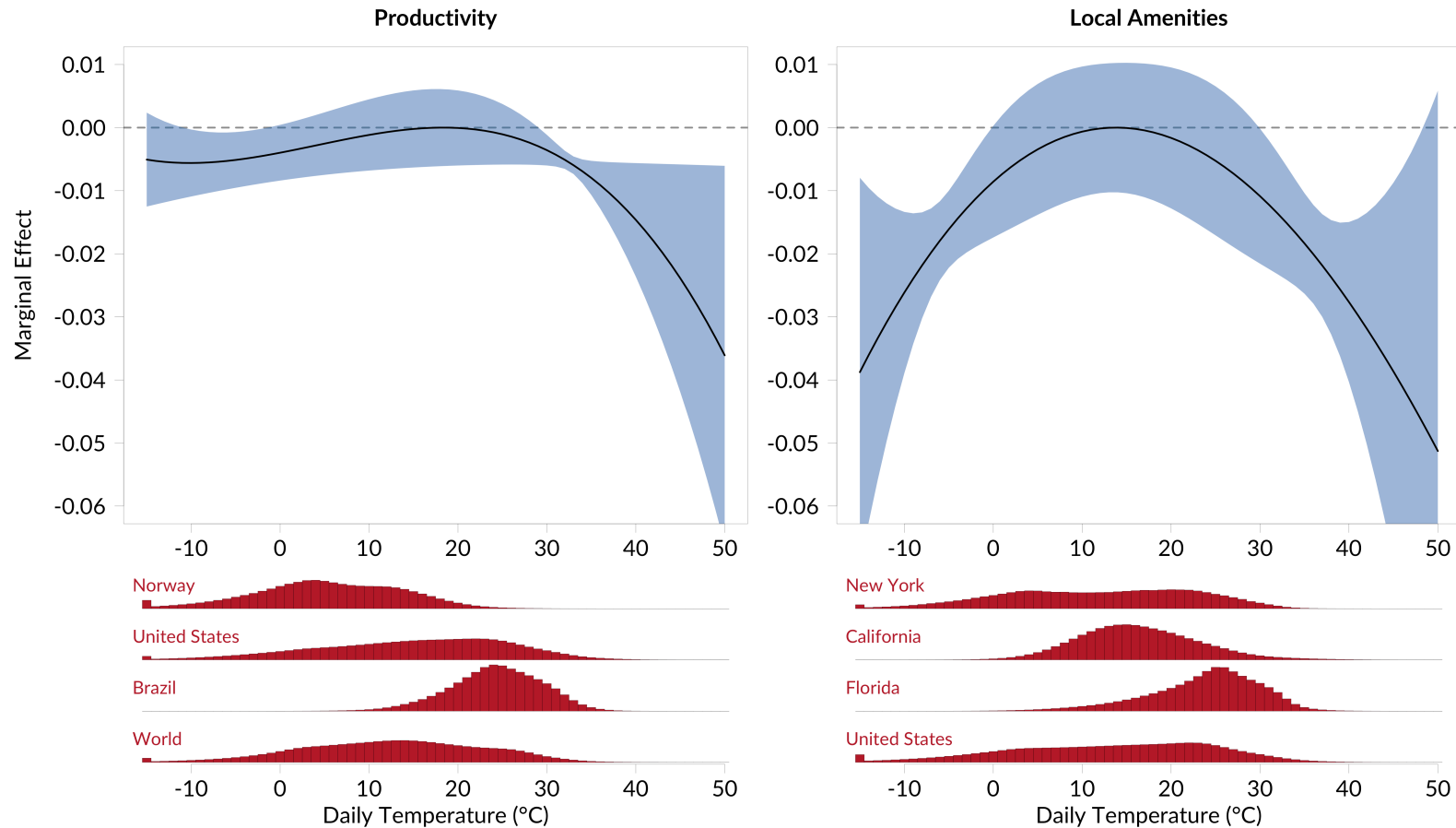
- Production
- Learning
- Leisure
- Fishing
- etc, etc

Climate change: heat hurts learning



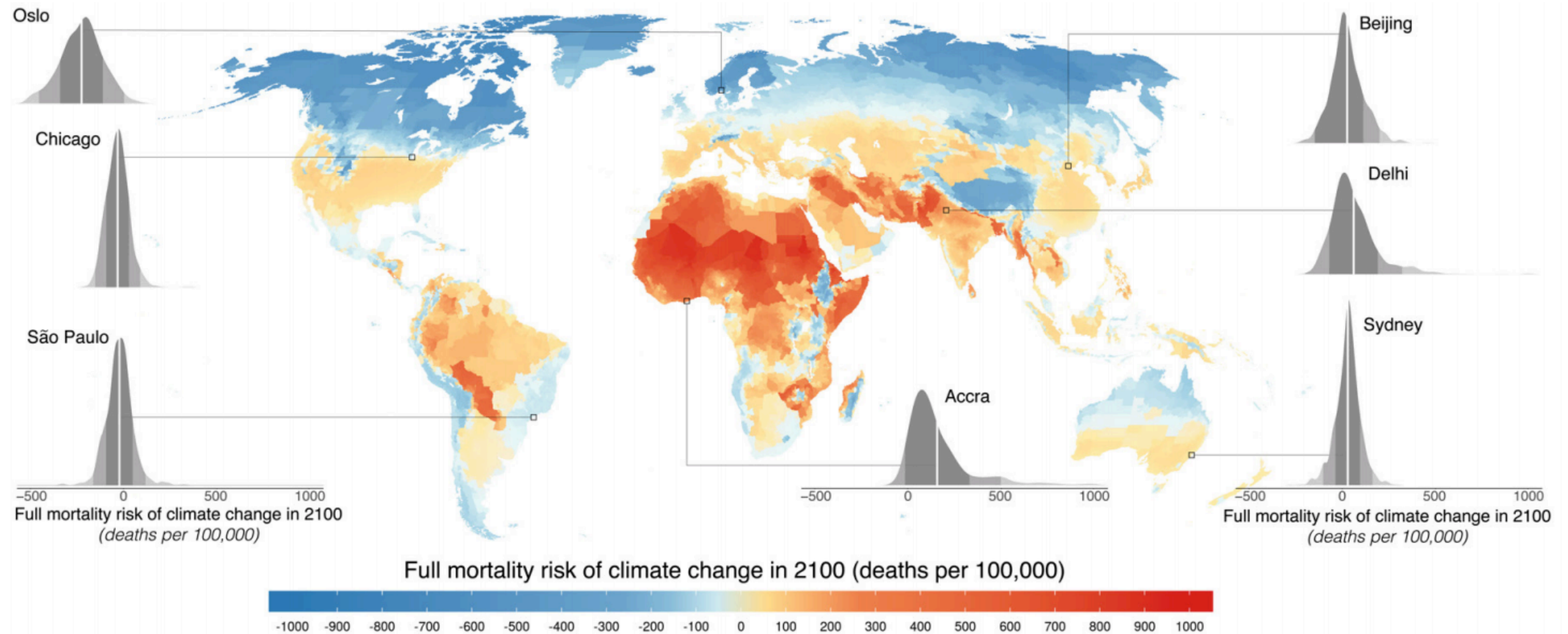
Park et al. (2020)

Climate change: heat hurts the economy



Rudik et al. (2021)

Climate change: extreme heat/cold increases mortality

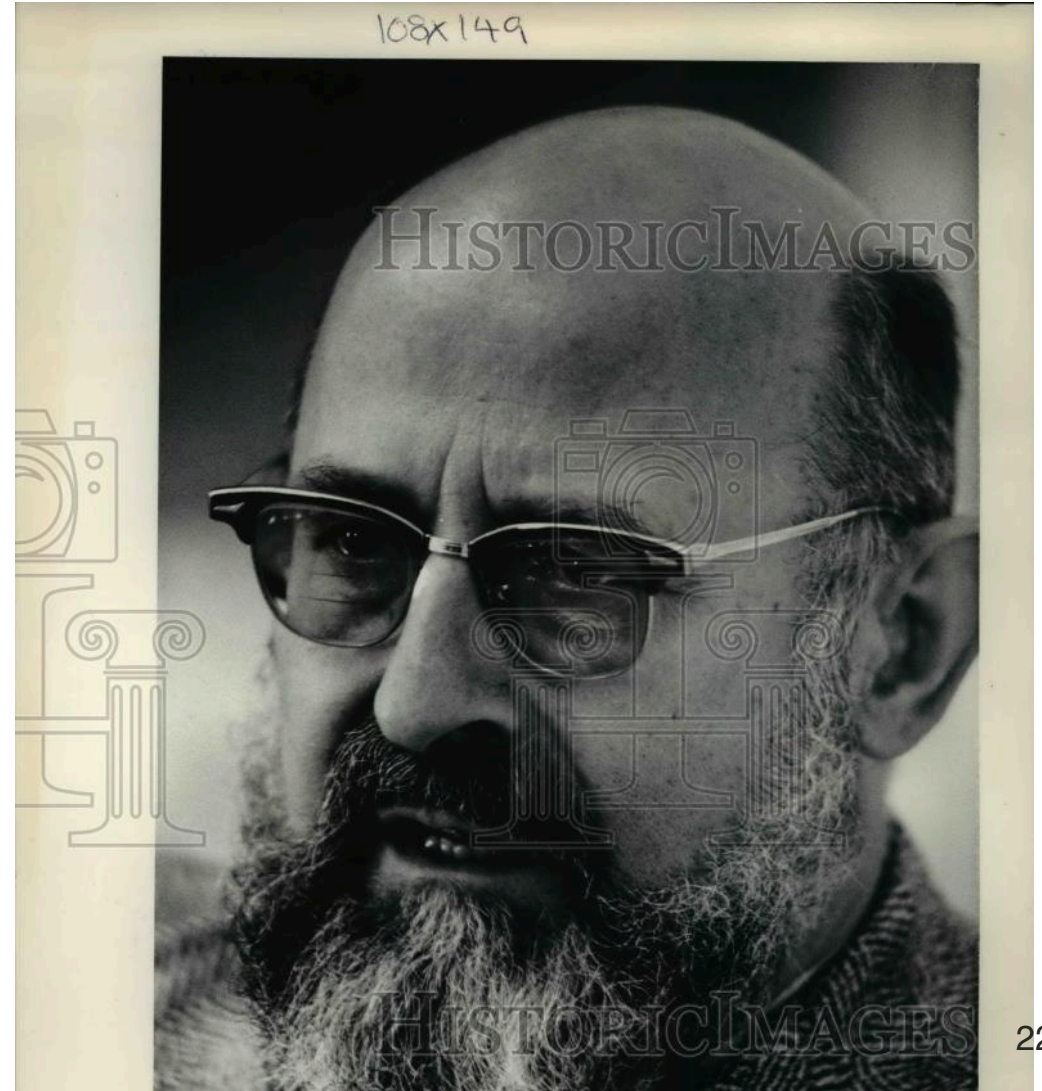


What is enviro econ?

Environmental economics is actually pretty new

Spurred by **John Krutilla** in the 1950s

His paper **Conservation Reconsidered** is the landmark paper in the field (sort of like *Wealth of Nations* and economics as a whole)



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Enhance your understanding of how economists think about solving real environmental problems

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Prepare you for after college:

- Industry jobs (energy, transportation, finance)
- Public sector and NGOs (EPA, DOE, RFF, Brookings, Federal Reserve)
- Graduate programs

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Learn both the theory and **applications** of environmental economics

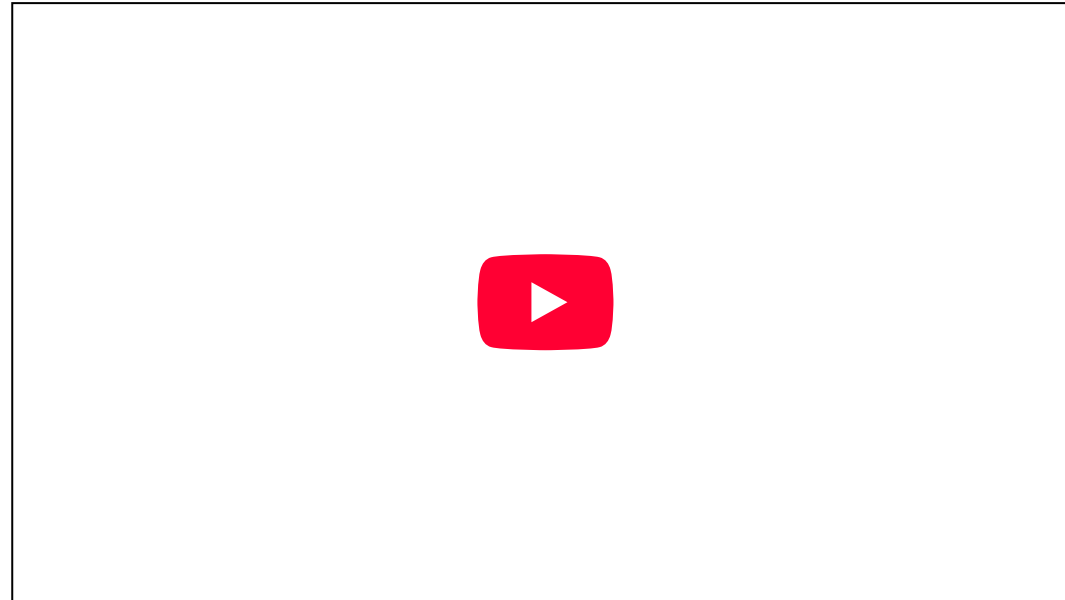
Microeconomics recap

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Is Intro to Micro applicable everywhere?

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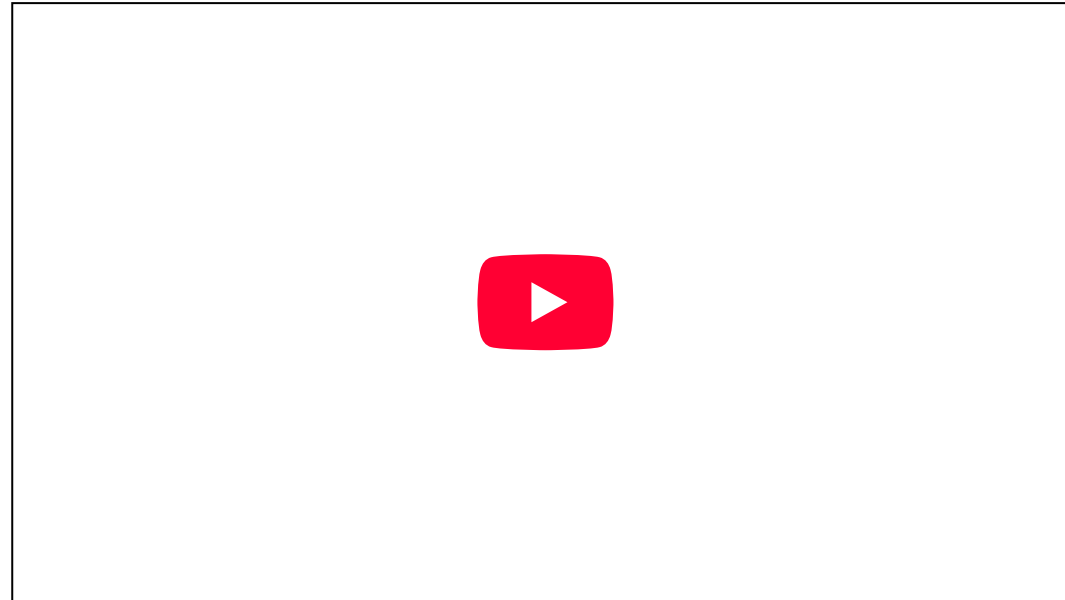


Creating markets to solve problems

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Micro recap: terminology

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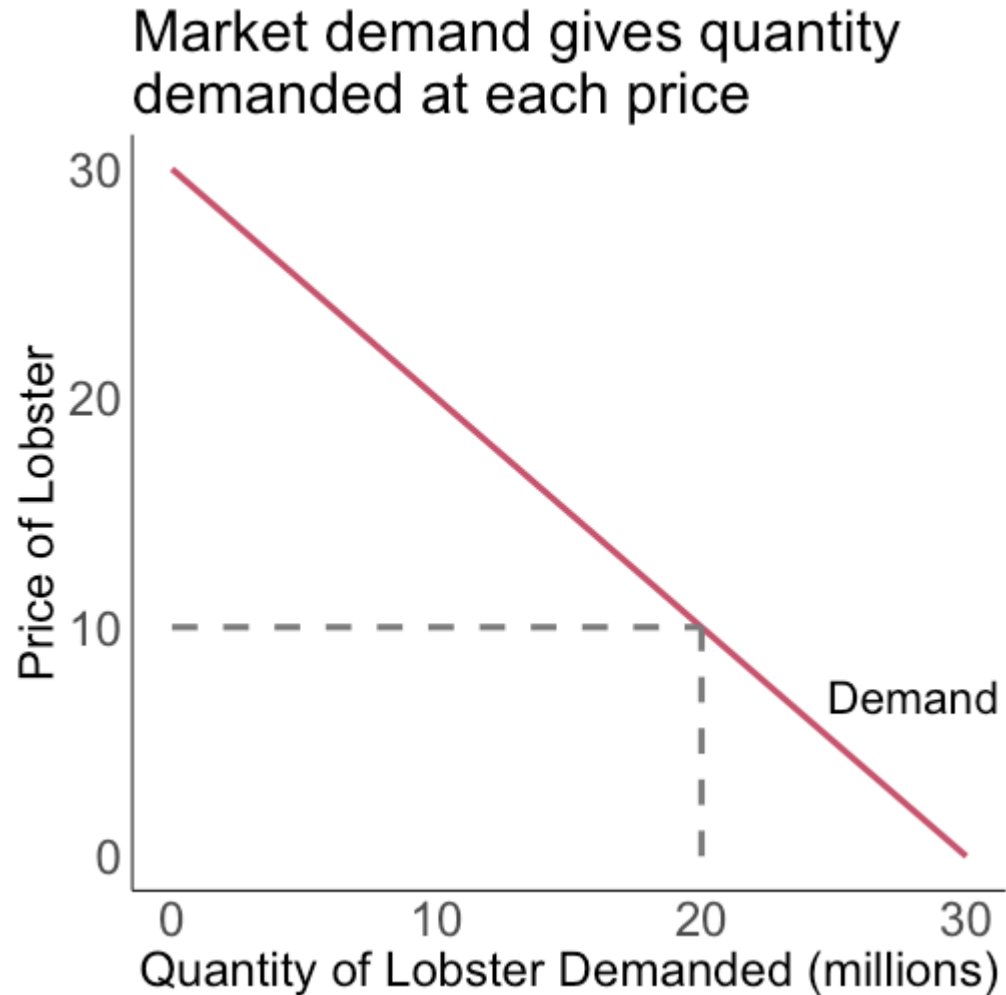
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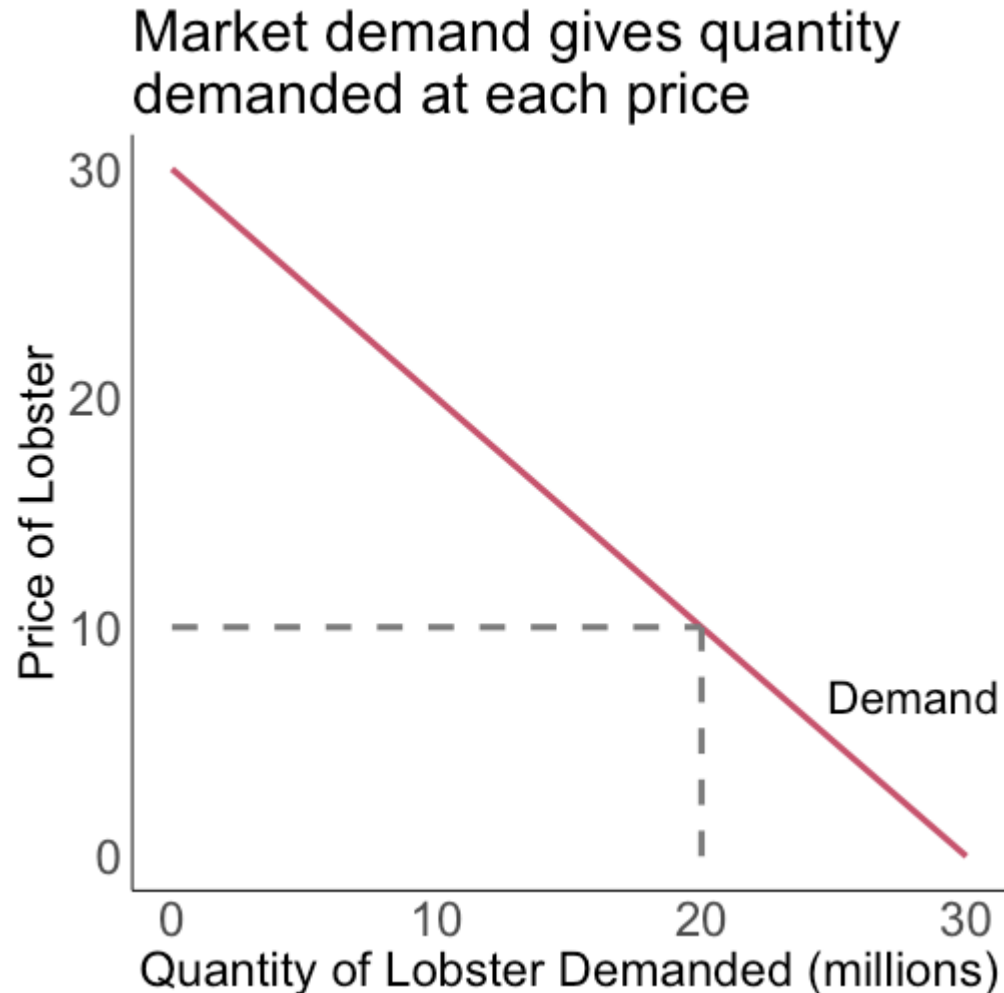
Supply curve: A schedule or graph showing the quantity of a good that sellers wish to sell at each price; it gives us the marginal willingness to accept or the marginal cost

Market demand



Market demand is aggregated from all individual demand curves

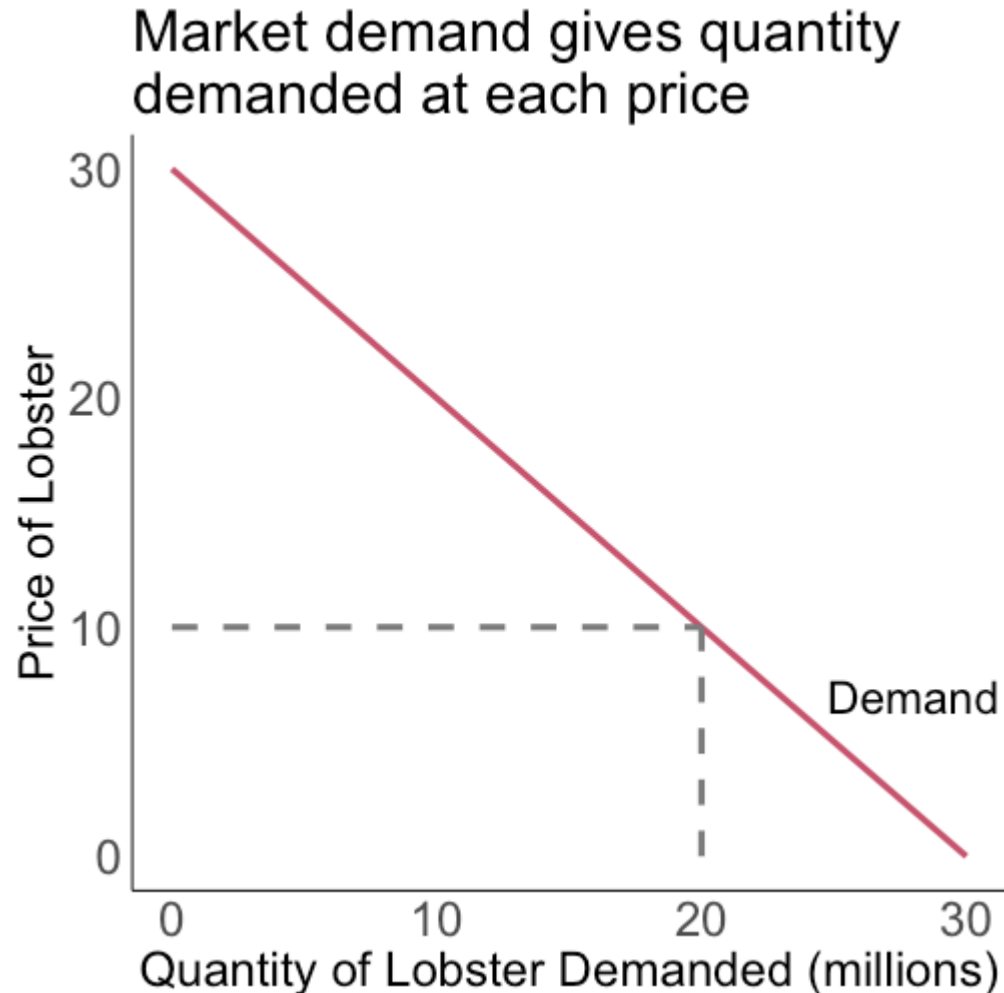
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Horizontal interpretation: if buyers face a price of \$10/lobster they will want to purchase 20 million

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Vertical interpretation: if buyers are buying 20 million lobsters, the marginal buyer is willing to pay at most \$10

Market demand slopes down

Why do demand curves slope down?

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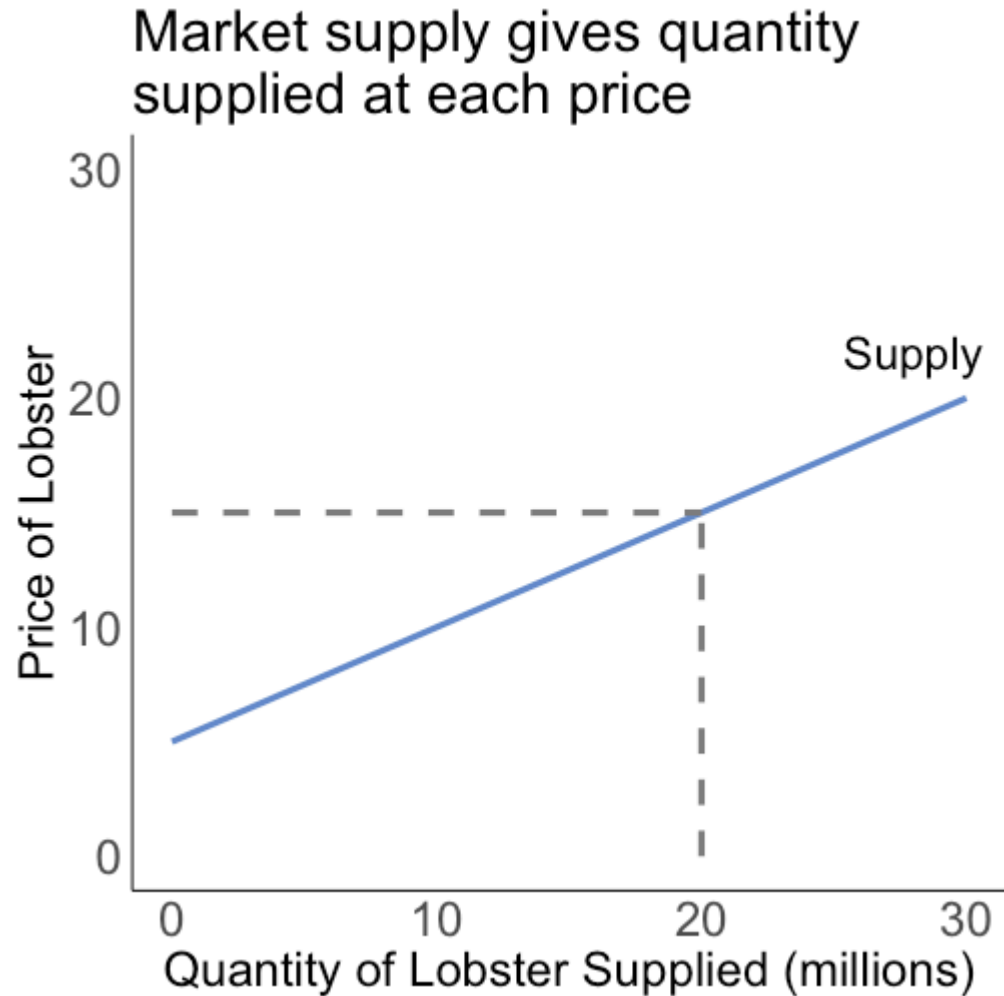
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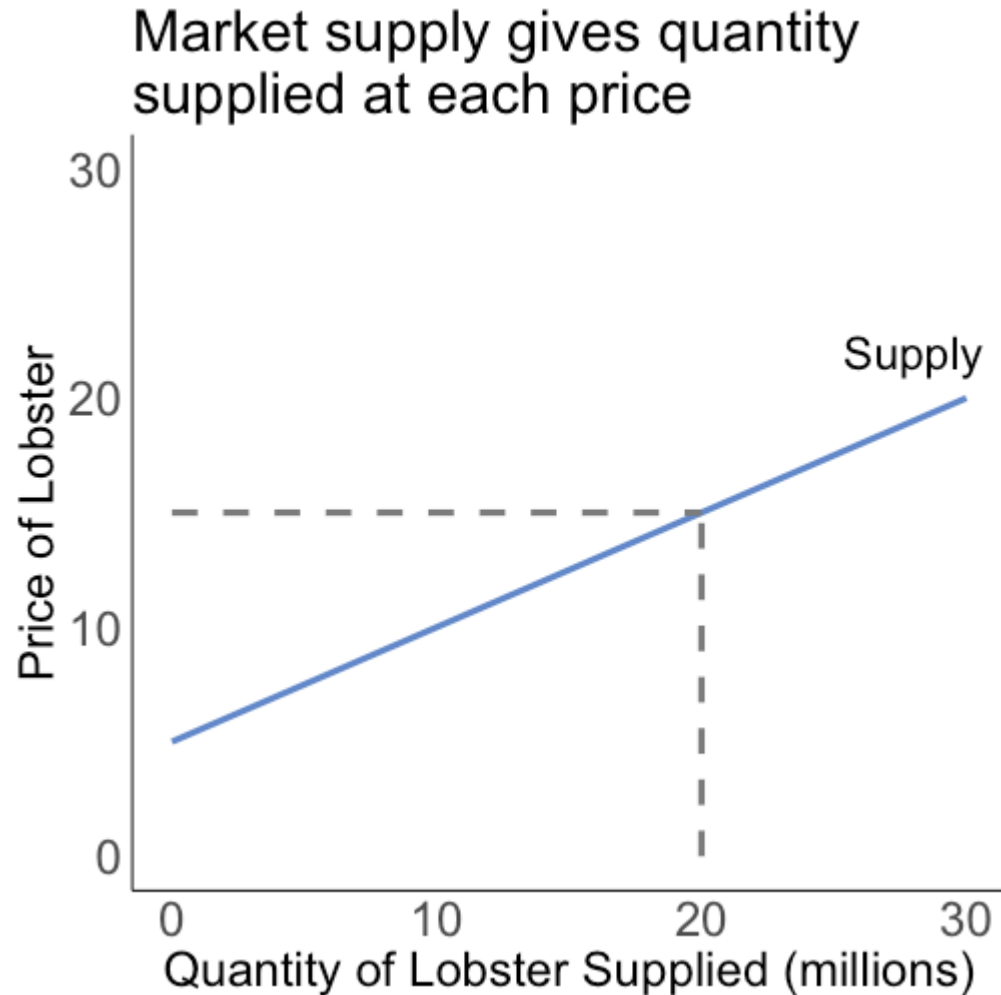
This is the **income effect**: if the price of pizza goes up, we have a lower real budget

Market supply



Market supply is aggregated from all individual supply/MC curves

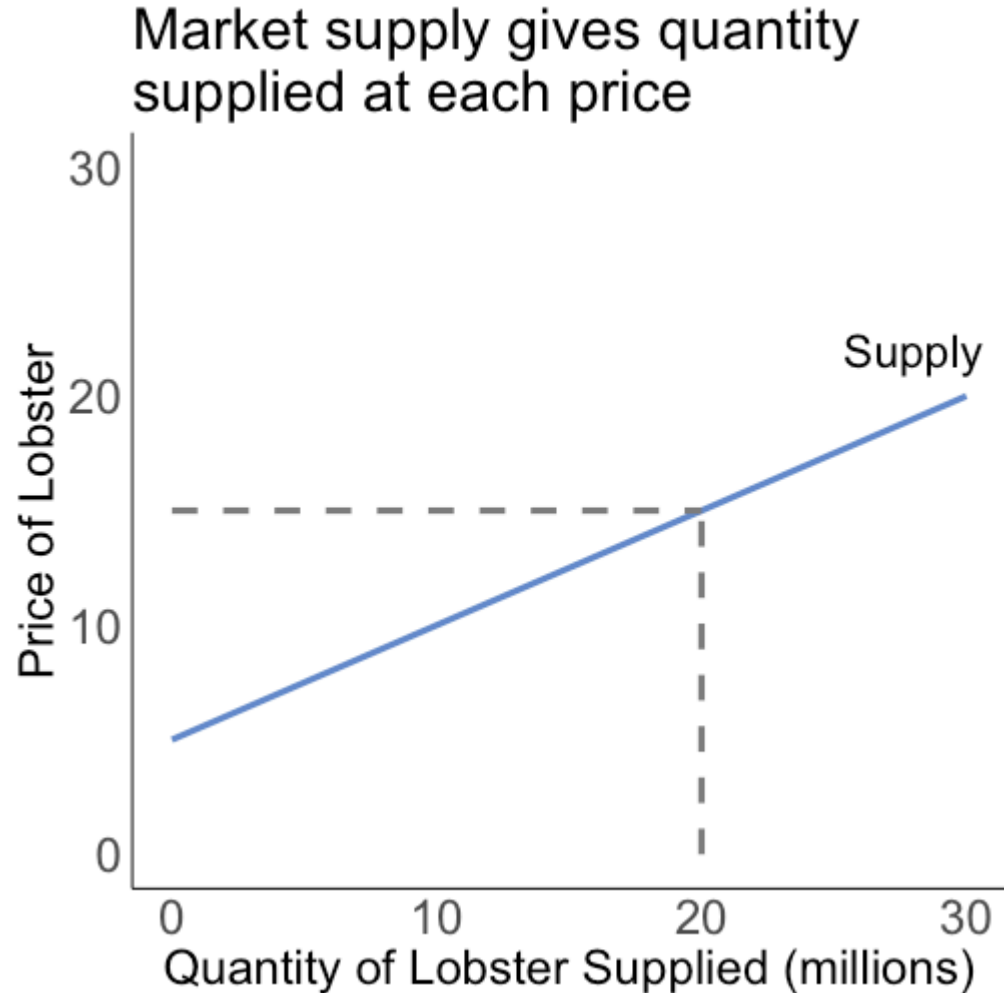
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Horizontal interpretation: if sellers face a price of \$15/lobster they will want to sell 20 million

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Vertical interpretation: if sellers are selling 20 million lobsters, the marginal cost of the last lobster is \$15

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If we produce more of a good, we choose the lowest (opportunity) cost production processes first, higher cost production processes later

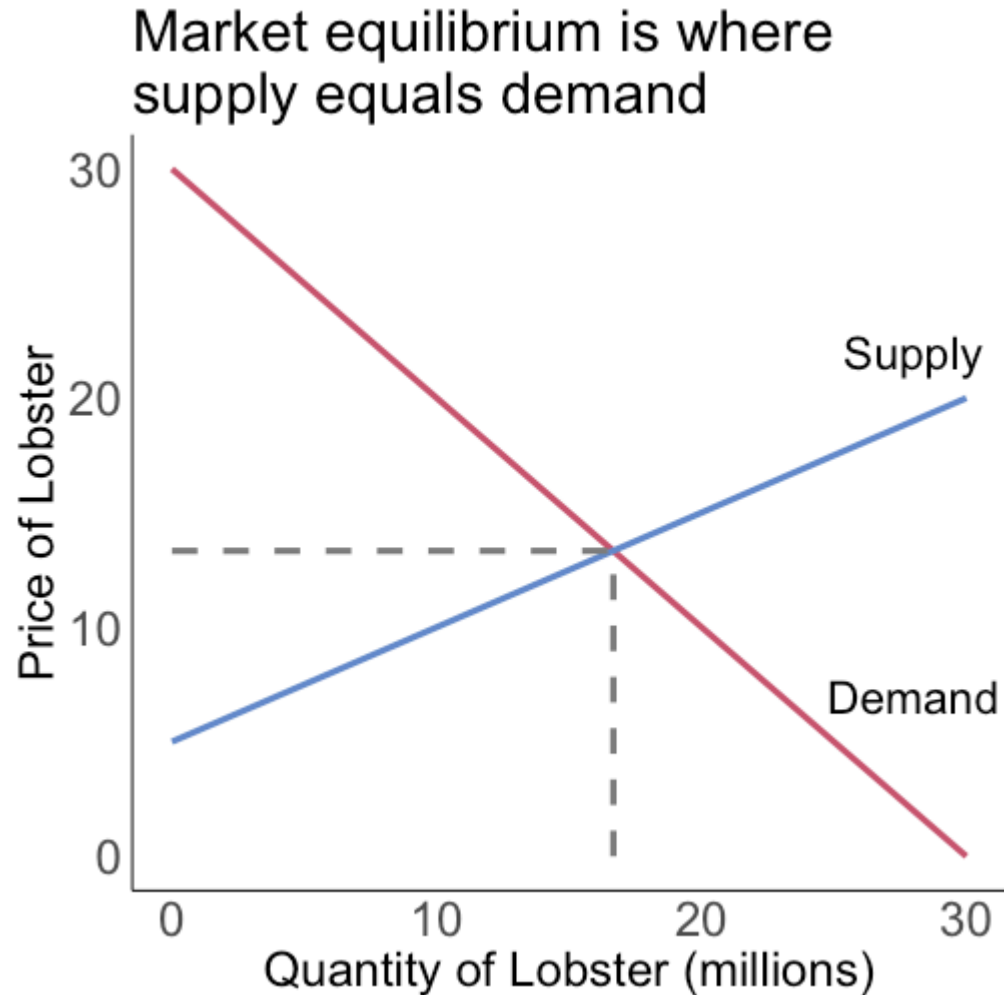
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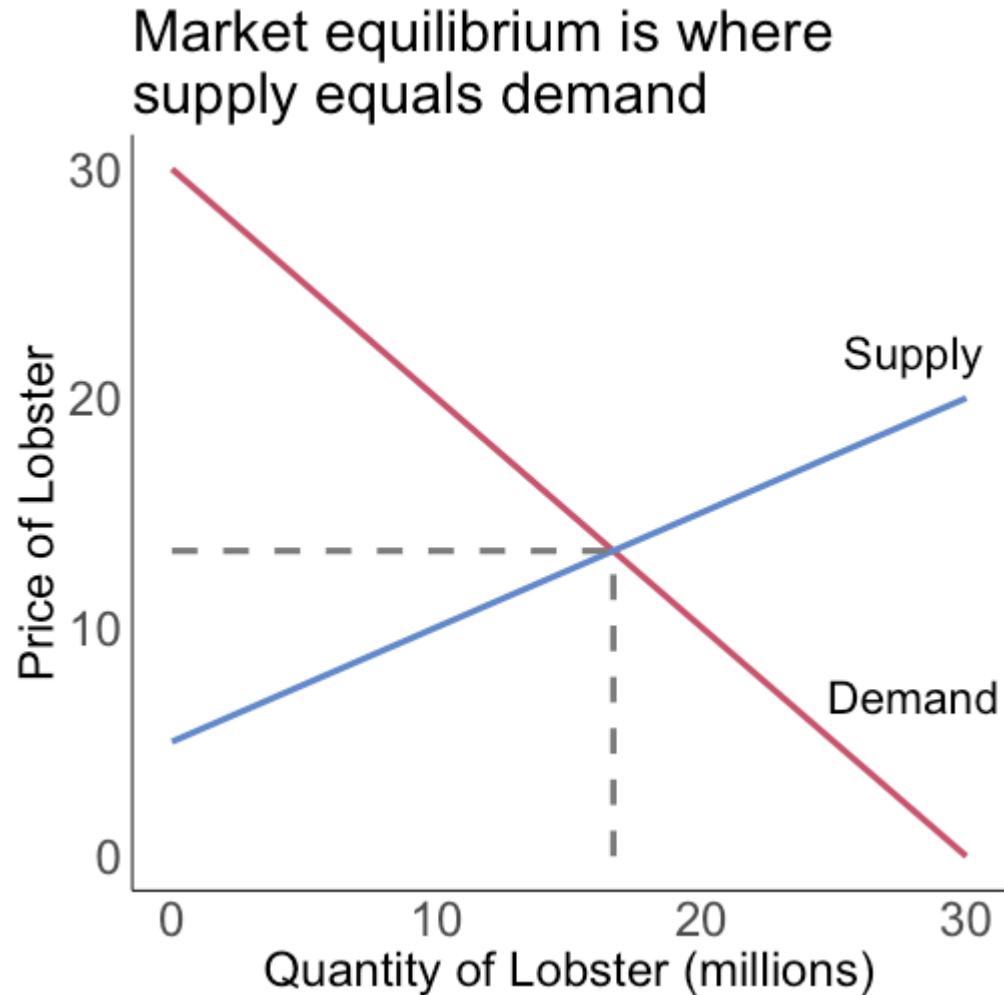
Marginal costs go up as production goes up → producers need higher prices in order to produce more goods

Market equilibrium



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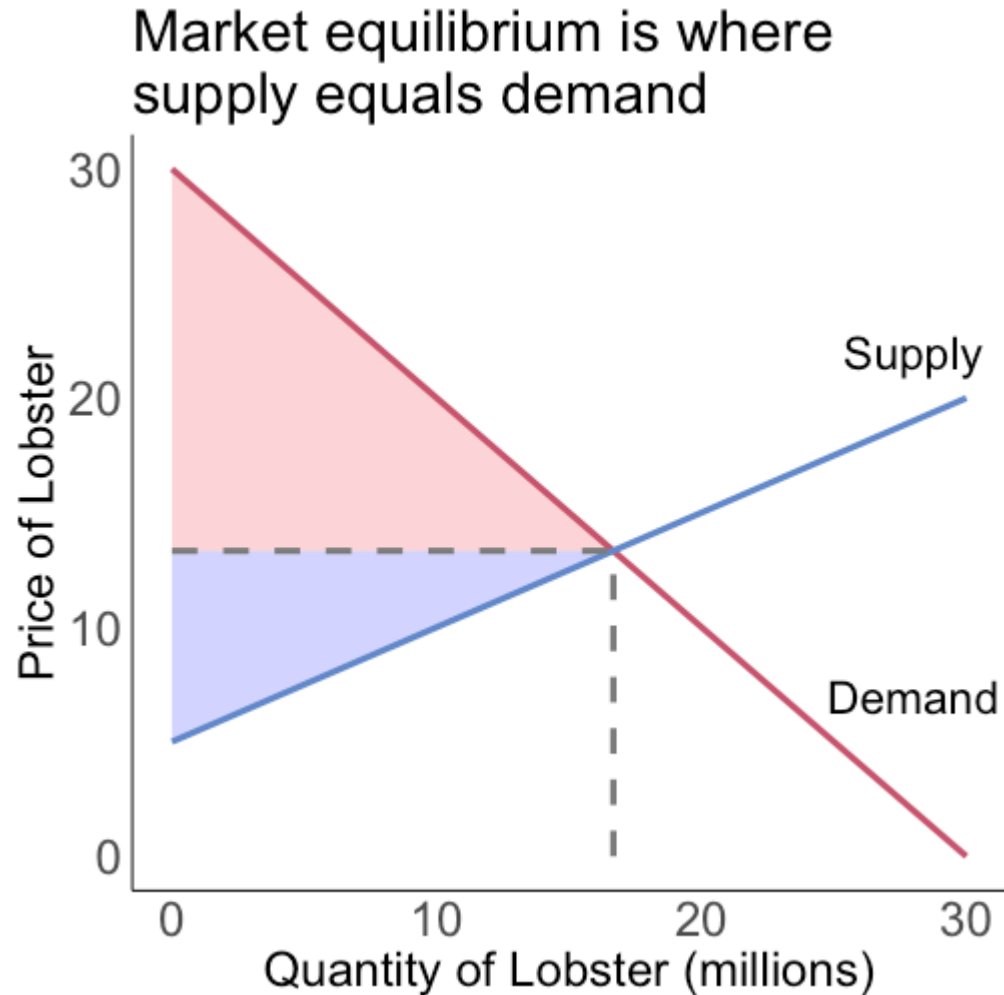
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This gives us the price where the quantity demanded exactly equals the quantity supplied: no shortages, no surpluses

Market equilibrium



Consumer surplus is the difference between willingness to pay (demand) and price

Producer surplus is the difference between price and marginal cost (supply)

Total surplus is the sum of CS and PS

Efficiency

We have two notions of efficiency:

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If you can make one person better off without making anyone else worse off, it's called a **Pareto improvement**

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Main takeaway: markets are often a nice way to allocate scarce resources

Under what assumptions are markets efficient?

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3. Complete markets (minimal transactions costs)
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They drive a wedge between private and social marginal cost, or private and social marginal benefit

The key departures in environmental economics

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	Excludable	Non-Excludable
Rival	Private goods (food)	Common-pool resources (fish, timber)
Non-rival	Club goods (parks, netflix)	Public goods (air, national defense)

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Why?

Without the adequate information, buyers or sellers cannot make the choices in their best interest

Efficiency and equity

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- If Elon Musk held **all** the wealth in the world, that would be a Pareto efficient outcome even though it goes against basically all people's notions of equity
- Perfect price discrimination is also Pareto efficient: producers capture all the surplus but it is maximized

Ecological wealth

